Alberta Public Health Disease Management Guidelines

Hantavirus

This publication is available online at https://open.alberta.ca/publications/hantavirus-pulmonary-syndrome

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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Clinical illness[(A)] with laboratory confirmation of infection:

- Detection of hantavirus-specific IgM antibodies;

OR

- Seroconversion or significant difference in hantavirus-specific IgG antibody titres taken 7–10 days apart;

OR

- Detection of hantavirus-specific nucleic acid (e.g., PCR) in an appropriate clinical specimen (e.g., blood)[(B)];

OR

- Detection of hantavirus antigen by immunostaining (pre- or post-mortem) in an appropriate specimen (e.g., blood, lung, kidney or spleen tissue)[(B)].

[(A)] Clinical illness is characterized by:

- A febrile illness (temperature > 38.3 °C oral) requiring supplemental oxygen

PLUS

- Bilateral diffuse infiltrates (may resemble acute respiratory distress syndrome [ARDS])

PLUS

- Develops within 72 hours of hospitalization in a previously healthy person

OR

- Unexplained respiratory illness resulting in death with an autopsy examination demonstrating non-cardiogenic pulmonary edema without an identifiable specific cause of death.

[(B)] Refer to the National Microbiology Laboratory Guide to Services for current specimen collection and submission information.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

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- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) **and** the Hantavirus Enhanced Surveillance Report (ESR) form of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) and ESR form within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days) including:
  - name,
  - date of birth,
  - out-of-province health care number,
  - out-of-province address and phone number,
  - positive laboratory report, and
  - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Hantavirus is a viral disease. There are multiple strains of hantavirus that cause different clinical illnesses. Several strains have been shown to cause hantavirus pulmonary syndrome (HPS). Approximately four different hantavirus species have been identified in North America and five in South America with serologic cross reactivity. The most common species is called Sin Nombre.

Clinical Presentation

Reference 2 applies to this section.

Hantavirus pulmonary syndrome (HPS) is a respiratory illness associated with the inhalation of aerosolized rodent excreta (urine and feces) contaminated by hantavirus particles, and contact with rodent saliva. Upon inhalation of hantavirus-contaminated particles, an extensive infection of pulmonary endothelial cells occurs and a viremic phase is initiated.

Following the incubation period, individuals typically present with very non-specific symptoms followed by a relatively short febrile prodrome lasting three to five days. Fever, chills, and myalgia as well as malaise, headaches and gastrointestinal symptoms are common symptoms. Five days after the onset of initial symptoms cough, shortness of breath, dizziness, sweats, and arthralgia may develop. Pulmonary edema and deterioration of cardiopulmonary function may rapidly occur over the next 24 hours. Typical findings include fever, tachypnea, and tachycardia. Once the cardiopulmonary phase begins, the disease progresses rapidly, necessitating hospitalization and often ventilation. The infection may be fatal.

Diagnosis

If clinical symptoms warrant and there is documented or suspected exposure, samples of serum and/or tissue (if present) are sent for confirmation to the National Microbiology Laboratory (Winnipeg) for antibody testing by ELISA (serum) and PCR on tissue. A clinical history form is completed by the requesting physician to provide documentation.

Treatment

- Supportive treatment.
- No specific antiviral drugs are currently used.

Reservoir

The major reservoir in North America is the deer mouse. Deer mice are found primarily in rural and semi-rural areas, often in old buildings and barns. Hantavirus antibodies have also been found in pack rats, chipmunks and other wild rodents.

Transmission

Hantavirus is primarily transmitted through inhalation of aerosolized rodent excreta (urine or feces) contaminated by hantavirus particles. When contaminated dried materials are disturbed they may be inhaled. Rarely they are introduced into broken skin, conjunctiva or ingested through contaminated food or water. Infection may also occur from rodent bites. There has been no evidence of person to person spread of virus species found in North America or transmission from pets or livestock to humans. One hantavirus species from South America (Andes virus) has shown evidence of person to person spread.

Infected rodents may shed the virus in urine, feces, and saliva for many weeks or months or for a lifetime. The greatest quantity of virus shed is approximately three to eight weeks after infection. Transmission from rodent to rodent is believed to be through physical contact.

Incubation Period

The incubation period is approximately two weeks, ranging from a few days to six weeks.

Period of Communicability

Person to person hantavirus transmission has only been reported from one species of hantavirus found in South America.

Host Susceptibility

All persons are presumed to be susceptible. Rural dwellers, cottagers and campers are most at risk. No asymptomatic infections have been documented, but milder infections without frank pulmonary edema have been observed. Protection and duration of immunity conferred by previous infection is unknown.

Incidence

General

Hantavirus infects rodents worldwide. HPS was first recognized in 1993 in the south-western United States of America (USA). Since then, sporadic cases have been recognized in 27 states in the USA and several provinces in Canada (British Columbia to Ontario). Recent studies in the USA indicate that infected deer mice are present in every habitat type, ranging from desert to alpine tundra.

HPS is more common in South America than in North America. There have been cases identified in Argentina, Chile, Uruguay, Paraguay, Brazil and Bolivia. In 1999, Panama experienced an outbreak of HPS following an episode of a severe flooding that resulted in an increase in the population of rodents. This represented the first cases of HPS in Central America. Other clinical syndromes (hemorrhagic fever and renal syndrome) are thought to be caused by other species of hantavirus found from Russia to South Korea.

Canada

Reference 2 applies to this section.

HPS was made a nationally notifiable disease January 1, 2000. The first Canadian case recognized during active surveillance occurred in 1994 in British Columbia. Subsequently, retrospective reports date back to the first case identified as early as 1989 in Alberta. There have been 32 lab-confirmed cases of hantavirus pulmonary syndrome in Canada from 1989 to 1999; six in British Columbia, 20 in Alberta, five in Saskatchewan and one in Manitoba. The case fatality rate is approximately 38%. The average age of HPS cases has been reported to be 39 years (range 15–62 years). The majority of cases (60%) have been male. Approximately half of the cases occurred during the months of April, May and June. The highest risk (70%) occupation was identified as farming. Disease occurrence coincides with the presence and increased number of carrier rodents.

Alberta

Reference 3 applies to this section.

Twenty eight cases have been reported in Alberta between 1989 and 2004 with a case fatality rate of 30%. The average age of the cases is 40 years. Most cases (18 cases) indicated obvious exposure to mice and or mice droppings/nesting material in the one to three week period prior to onset of illness. Of those 18 cases, 10 people also indicated that they lived or worked in a rural setting. Four cases reported mice in or around their property but that there was no obvious exposure to mice or mice droppings/nesting material. Five cases had no known exposure to mice.

Public Health Management

Key Investigation

- History of relevant exposure including exposure to mice, their saliva and their excrement (feces and urine); and in particular:
  - occupational exposure (e.g., farm or other outdoor worker),
  - exposure through camping, hiking, etc.,
  - living in a rural setting, and/or
  - cleaning or working in mouse infested areas.
- The PHAC Hantavirus Pulmonary Syndrome Surveillance Case Report is available from the CMOH and should be completed by the RHA and attending physician(3).
- Laboratory sampling from mice known to be associated with a case is not recommended. Periodic surveys of mouse populations throughout Alberta have been performed by Health Canada.

Management of a Case

- Consultation with an infectious diseases physician for investigation and management is recommended.
- Isolation precautions are not required, as person to person transmission has not been documented.

Management of Contacts

- Symptomatic and asymptomatic contacts should be investigated if a common source is suspected.

Preventative Measures

References 4 to 6 apply to this section.

- Educate the public, in particular farmers and other rural residents, to avoid exposure.
- Prevent mice from entering homes by using steel wool, metal roof flashing or cement to cover all openings from the outside.
- Use mouse proof storage containers for food, pet and animal food, grain and garbage.
- Trap and dispose of mice in settings where humans reside.
  - Dispose of dead mice, mice droppings and nests using plastic or rubber gloves
  - Wet carcasses, nests and/or droppings with a disinfectant such as dilute bleach (1 part bleach and 10 parts water) or household disinfectant.
  - Allow 10 minutes for the disinfectant to act.
  - Place article in double plastic bags or burn them.
- Discourage rodents from living around homes by:
  - keeping grass short,
  - storing hay on pallets,
  - placing wood piles 100 feet or more from the home and raising them at least 12 inches off the ground, and
  - removing abandoned vehicles, discarded tires and old, unused buildings (which may serve as nesting sites) from the property.
- Camping or outdoor activities can expose individuals to mice and their droppings. Avoid direct contact with areas where there is evidence of rodents (e.g., tunnels, nests, rodent feces).

- Educate the public (farmers and other rural residents) about cleaning areas contaminated by mouse nests and mouse urine, feces and saliva.
  - Ventilate the area by opening windows and doors for 30 minutes before and after disinfecting.
  - Water down the area with disinfectant such as dilute bleach, (1 part bleach and 10 parts water).
  - If area is carpeted use a mixture of water, detergent, and other household disinfectant (e.g., Lysol) or commercially steam clean area.
  - Remove droppings by damp mopping, preferably twice. Mouse droppings should not be removed by sweeping or vacuuming.
  - Wear full length clothing to avoid skin contamination.
  - Use gloves to handle soiled clothes.
  - Wash dirty laundry with hot water and detergent, and dry thoroughly.
  - If dust generation cannot be avoided seek professional help.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
October 2021	General	• Updated Template • Diagnosis and Treatment section moved to Epidemiology • Checked web links

References

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